



CSA1511 – CLOUD COMPUTING AND BIG DATA ANALYTICS

CO1 – ASSESSMENT TOOL – 3

NAME : NAVADEEP R

REG NO : 192424110

FACULTY INCHARGE : Dr.HEMAVATHI R

1. Define Cloud Computing and Explain Any Four Characteristics that Make it Different from Traditional On-Premise Computing

Definition

Cloud Computing is a technology that provides computing resources such as servers, storage, databases, networking, software, and processing power over the Internet on an on-demand basis. Instead of purchasing and maintaining physical hardware, users can access these resources whenever needed and pay only for what they use.

Traditional on-premise computing requires organizations to purchase, install, manage, and maintain their own servers and infrastructure within their premises, whereas cloud computing delivers these resources as online services.

Characteristics of Cloud Computing

1. On-Demand Self-Service

Users can provision computing resources such as virtual machines, storage, and databases whenever required without interacting directly with the cloud service provider.

Example:

A developer can instantly create a virtual server on Amazon Web Services (AWS) within a few minutes.

2. Broad Network Access

Cloud services are available over the Internet and can be accessed from different devices such as laptops, smartphones, tablets, and desktops.

Example:

Google Drive allows users to access their files from anywhere using any internet-connected device.

3. Resource Pooling

Cloud providers maintain a shared pool of computing resources that are allocated dynamically to multiple users using virtualization technologies.

Example:

Microsoft Azure shares its physical servers among thousands of customers while keeping each customer's data isolated.

4. Rapid Elasticity (Scalability)

Cloud resources can be increased or decreased automatically according to user demand.

Example:

Netflix increases server capacity during weekends when millions of users stream movies simultaneously.

Difference from Traditional On-Premise Computing

Cloud Computing	Traditional On-Premise Computing
Resources available through Internet	Resources located inside organization
Pay-as-you-use model	High initial investment
Easily scalable	Scaling requires purchasing new hardware
Managed by cloud provider	Managed by organization

2. Trace the Evolution of Cloud Computing from Hardware Evolution to Internet Software Evolution, and Show How These Changes Enabled Modern Cloud Platforms

Cloud computing evolved gradually through several technological advancements.

Stage 1: Hardware Evolution (1960s–1980s)

Initially, organizations used large mainframe computers shared among multiple users through terminals.

Characteristics

- Centralized computing
- Expensive hardware
- Limited accessibility

Limitation

Only large organizations could afford these systems.

Stage 2: Personal Computer Era (1980s–1990s)

Personal computers became affordable and users started running applications locally.

Characteristics

- Individual ownership
- Standalone applications
- Local storage

Limitation

Data sharing and collaboration were difficult.

Stage 3: Networking and Internet Evolution (1990s–2000s)

The widespread adoption of the Internet enabled communication between computers across the world.

Improvements

- Email
- Web applications
- Remote access
- File sharing

Businesses began hosting applications online.

Stage 4: Virtualization Technology

Virtualization allowed multiple virtual machines to run on a single physical server.

Benefits

- Better hardware utilization
- Reduced operational cost
- Improved scalability
- Easier resource management

This became the foundation of cloud computing.

Stage 5: Modern Cloud Computing

Companies such as AWS, Microsoft Azure, and Google Cloud introduced cloud platforms providing computing resources over the Internet.

Modern cloud platforms offer:

- Virtual Machines
- Cloud Storage
- Artificial Intelligence
- Big Data Analytics
- Serverless Computing
- Machine Learning

Evolution Flow

Hardware Evolution



Mainframe Computing



Personal Computers



Networking & Internet



Virtualization



Cloud Computing



Modern Cloud Platforms

3. Explain the Role of Server Virtualization in Cloud Computing. Differentiate Between Virtualization as an Enabling Technology and Cloud Computing as a Service Model

Server Virtualization

Server virtualization is the process of creating multiple virtual servers (Virtual Machines) on a single physical server using software called a **Hypervisor**.

Each virtual machine behaves like an independent computer with its own operating system, memory, CPU, and storage.

Role of Server Virtualization

Virtualization is the core technology that makes cloud computing possible.

It provides:

- Efficient hardware utilization
- Multiple operating systems on one server
- Reduced infrastructure cost
- Easier backup and recovery
- Improved scalability
- Isolation between applications

Example

One physical server can run:

- Windows Server VM
- Ubuntu Linux VM
- Red Hat Linux VM

simultaneously without interference.

Difference Between Virtualization and Cloud Computing

Virtualization	Cloud Computing
Technology	Service Model
Creates Virtual Machines	Delivers computing services
Runs on physical hardware	Runs on virtualized infrastructure
Uses Hypervisor	Uses Internet
Improves hardware utilization	Provides scalable services to users

Relationship

Virtualization is the enabling technology.

Cloud Computing is the service delivered using virtualization.

Without virtualization, cloud providers cannot efficiently share hardware among multiple customers.

4. Discuss the Role of Data Centers in Delivering Cloud Services

Definition

A Data Center is a secure facility that contains thousands of servers, storage systems, networking devices, cooling systems, and power infrastructure used to deliver cloud services.

Cloud providers operate multiple data centers worldwide.

Components of a Data Center

- Servers
- Storage Systems
- Networking Equipment
- Routers and Switches
- Backup Systems
- Cooling Systems
- Fire Protection
- Security Systems
- Power Supply

Role of Data Centers

1. Data Storage

Stores customer files, databases, and applications.

Example: Google Drive stores files in Google's data centers.

2. Computing Power

Provides CPU and memory resources for running applications.

3. High Availability

Multiple backup systems ensure services remain available even during hardware failures.

4. Security

Provides physical and network security including:

- CCTV surveillance
- Firewalls
- Encryption
- Access control

5. Disaster Recovery

Maintains backup copies of data across multiple geographical locations.

6. Scalability

Allows cloud providers to quickly increase computing resources based on customer demand.

Real-World Example

When a user uploads a document to Google Drive:

User Device



Internet



Google Data Center



Storage Server



Data Backup



User can access the file anytime from anywhere.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with One Suitable Real-World Example for Each Model

Cloud computing services are categorized into different service models based on the level of control provided to users.

Infrastructure as a Service (IaaS)

Definition

IaaS provides virtualized computing infrastructure such as virtual machines, storage, networking, and servers over the Internet.

The cloud provider manages the physical infrastructure, while users manage the operating system and applications.

Example

Amazon EC2 (AWS)

Use Case

A software company creates virtual servers to host its web applications.

Platform as a Service (PaaS)

Definition

PaaS provides a complete development environment where developers can build, test, and deploy applications without managing servers.

Example

Google App Engine

Use Case

Developers deploy web applications directly without worrying about server configuration.

Software as a Service (SaaS)

Definition

SaaS delivers ready-to-use software applications over the Internet.

Users simply access the software through a web browser.

Example

Google Docs

Use Case

Students create assignments online without installing Microsoft Office.

Everything as a Service (XaaS)

Definition

XaaS refers to delivering any IT service through the cloud, including infrastructure, software, databases, networking, security, storage, and artificial intelligence.

Example

Microsoft Azure

Use Case

An organization uses Azure for virtual machines, databases, AI services, storage, networking, and security from a single cloud platform.

Comparison of Cloud Service Models

Service Model	Provider Manages	User Manages	Real-World Example
IaaS	Hardware, Networking, Storage	Operating System, Applications, Data	Amazon EC2 (AWS)
PaaS	Infrastructure, Operating System, Runtime	Applications and Data	Google App Engine
SaaS	Everything	Only uses the software	Google Docs
XaaS	Delivers multiple cloud services	Uses required services	Microsoft Azure

Conclusion

Cloud computing has transformed the way organizations use computing resources by providing scalable, cost-effective, and on-demand services over the Internet. Technologies such as server virtualization and modern data centers form the foundation of cloud platforms. Different service models—including IaaS, PaaS, SaaS, and

XaaS—allow users to choose the level of control and management that best fits their needs. Together, these technologies enable reliable, flexible, and efficient computing for businesses, educational institutions, and individual users.